

A FIRST MODEL

Can years of experience predict salary?

A gentle introduction to linear regression.



An illustrative relationship, not a promise.

START WITH THE QUESTION

The question is simple, but salary is not

A model can begin with one input while the real world keeps many.

01

Input

Years of experience

02

Output

Predicted salary

03

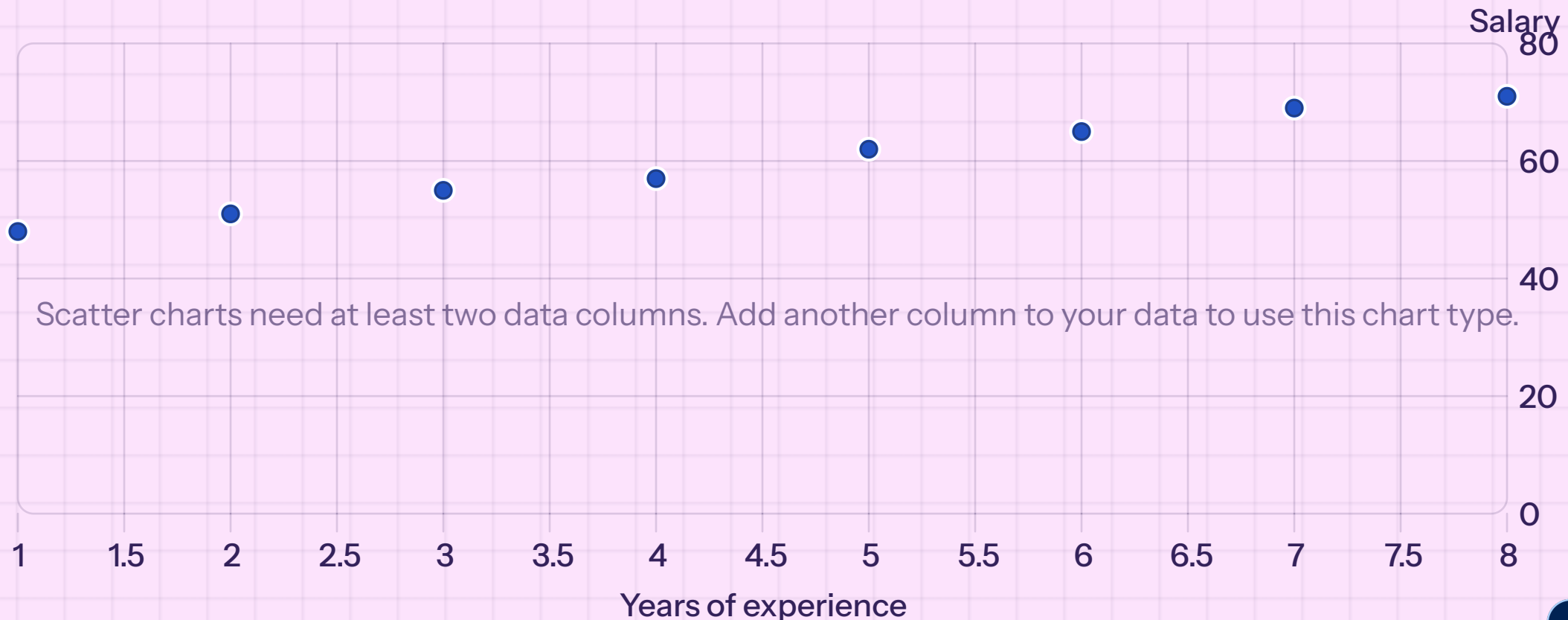
Reality check

Education, role, location, industry, and negotiation also matter.

THE BASIC IDEA

Linear regression turns examples into a line

Use past observations to estimate a relationship between one predictor and a continuous outcome.



The dots are observations. The rising line summarizes their overall direction.

READING THE LINE

The equation has two ideas

Together, they turn experience into a predicted salary.

01 • INTERCEPT

the model's
starting point

Where the line begins
when experience is zero.

02 • SLOPE

the predicted
salary change

How much the prediction
changes for one
additional year of
experience.

$$\text{salary} = \text{intercept} + \text{slope} \times \text{experience}$$

Ordinary least squares chooses coefficients that minimize the total squared residual error.

WORKED EXAMPLE

A worked example makes the line tangible

Suppose the fitted model is:

predicted salary
= \$45,000 +
\$3,500 × years
of experience.

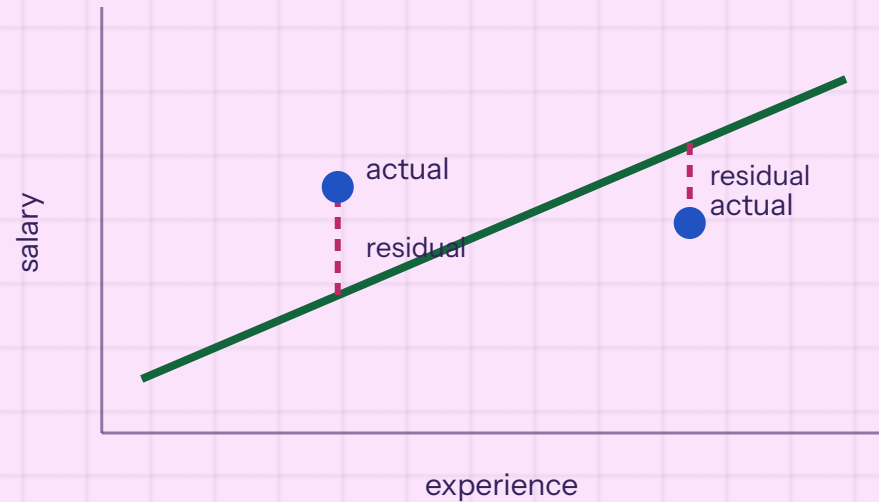
- 01 At 2 years: **\$52,000**
- 02 At 6 years: **\$66,000**
- 03 The slope is an average model relationship, not a guaranteed raise.

WHAT THE LINE MISSES

Every prediction has a residual

Residual = actual salary – predicted salary.

Points above the line are under-predicted; points below it are over-predicted.



The residual is the vertical gap between an observation and the line.

Accuracy is more than a convincing line

A line can look persuasive and still miss important parts of the data.



Hold-out data

Compare predictions with held-out data, not only the examples used to fit the line.



Mean absolute error

Use an error measure to express typical miss size.



R^2

Use it as a helpful summary of explained variation, not proof that the model is fair or complete.

THE RESPONSIBLE TAKEAWAY

Use the model as a starting point, not a verdict

Prediction supports judgment; it does not replace context.

01 Check the shape

Check whether the relationship is roughly linear.

02 Look for exceptions

Watch for outliers and groups the model treats differently.

03 Add context

Add relevant features before making high-stakes decisions.

04 Keep human judgment